

Patterns of condemnation rates in swine from a federally inspected abattoir in relation to disease outbreak information in Ontario (2005–2007)

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Abstract — Strong correlations between clinical signs on farms and the presence of lesions at slaughter have been reported. The objective of this study was to determine if changes in condemnation rates provide a data source for surveillance of disease outbreaks in pigs. The data were obtained from 1 abattoir in Ontario (2005–2007). The epidemiological relevance of the results was based on an outbreak of porcine circovirus associated disease (PCVAD) in Ontario in 2005. The total condemnations and condemnations due to arthritis and pneumonia patterns reflected the field infection of PCVAD in 2005 followed by the widespread use of porcine circovirus type 2 (PCV-2) vaccine in 2007. In contrast, increased rates of nephritis and enteritis suggested areas for enhanced surveillance for unexplained changes in disease patterns not identified through traditional passive surveillance. Further studies looking at the benefits of using abattoir data should compare condemnation patterns with multiple sources of swine health data.

Résumé — Schémas des taux de réforme chez les porcs dans un abattoir inspecté par le gouvernement fédéral en rapport avec l'information sur l'éclosion de maladies en Ontario (de 2005 à 2007). De fortes corrélations entre les signes cliniques à la ferme et la présence de lésions lors de l'abattage ont été signalées. L'objectif de la présente étude consistait à déterminer si les changements des taux de réforme fournissent une source de données pour la surveillance des éclosions de maladie chez les porcs. Les données ont été obtenues dans 1 abattoir en Ontario (de 2005 à 2007). La pertinence épidémiologique des résultats s'est fondée sur une éclosion de maladie associée au circovirus porcin (MACVP) en Ontario en 2005. Le total des réformes et les réformes attribuables aux patrons d'arthrite et de pneumonie ont reflété l'infection sur le terrain par le MACVP en 2005, suivie de l'usage généralisé du vaccin contre le circovirus porcin de type 2 (PCV-2) en 2007. Par contraste, les taux accrus de néphrite et d'entérite ont suggéré des domaines de surveillance accrue pour des changements inexpliqués au niveau des patrons de maladies non identifiés lors de la surveillance passive traditionnelle. De nouvelles études examinant les avantages de l'utilisation des données de l'abattoir devraient comparer les patrons de réforme avec des sources multiples de données sur la santé porcine.

(Traduit par Isabelle Vallières)

Can Vet J 2011;92:35–42

Introduction

Reliable surveillance systems may provide early warning to identify outbreaks of disease in animal populations (1,2). Syndromic surveillance focuses on clinical signs or lesions rather

than specific diagnoses and has become increasingly important as it may identify natural disease outbreaks more quickly than surveillance systems based on laboratory diagnoses (3–5). Syndromic information can be obtained from farms, veterinary clinics, diagnostic laboratories, and abattoirs. The validity of the classification of lesions at abattoirs has been questioned because of the difficult conditions under which diagnostic work is carried out during meat inspection (6) and the variability of classification of lesions in different abattoirs (7). Detection of changes in condemnation rates has been used in some epidemiological studies to evaluate changes in the health status of pig populations so control strategies of common infectious diseases could be evaluated (1,6–10). Correlations between clinical signs at the farm level and lesions at slaughter have been reported for pneumonia and diarrhea and other complex diseases (6,11). Willeberg et al (6) reported highly significant associations between chronic respiratory lesions at slaughter

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and pigs with clinical signs of respiratory disease on-farm ($RR = 3.5$, $P < 0.001$). In addition, a strong correlation was observed between specific lung infections prior to slaughter and postmortem diagnoses of particular diseases affecting the animal (6). However, the usefulness of slaughter data as a source of surveillance data to detect specific outbreaks of disease has not been evaluated.

In Ontario, swine disease outbreaks in finisher herds associated with swine influenza virus (SIV), porcine reproductive and respiratory syndrome virus (PRRSv), and porcine circovirus type 2 (PCV-2) occurred in early 2005 (12–14). In particular, the expression of PCVAD was different than previously experienced with older animals affected and higher reported mortality rates (15). Affected pigs had rasping coughs and watery diarrhea. Postmortem findings included various degrees of broncho-interstitial pneumonia and thickening of the jejunum, ileum, and colon. Affected animals also had lesions associated with interstitial nephritis (15,16). This change in the expression of PCVAD coincided with a change in the predominant viral restricted fragment length polymorphism (RFLP) pattern from 422 to 321 (16). The number of cases typed as RFLP type 321 increased from 1 in 2004 to 135 out of 175 cases typed in 2005 by the Animal Health Laboratory (AHL) at the University of Guelph, Guelph, Ontario. In 2006, 130 out of 159, and in 2007 53 out of 58 typed cases were associated with RFLP 321 (17). A total of 350 PCVAD cases were presented to the AHL in 2005 and 408 in 2006; however, only 222 new PCVAD cases were reported in 2007 (17). Between 2005 and 2006, the percentage of total submissions diagnosed with PCV-2 increased from 8.9% to 10.1% (18–20); however, the percent of PCVAD cases compared to the total swine submissions decreased to 6.3% in 2007 (16), when sufficient quantities of PCV-2 vaccine became available to producers (21).

A wide spectrum of infectious agents was observed concomitant with PCV-2 on affected farms during the outbreak. These include *Mycoplasma hyopneumoniae*, *Mycoplasma hyorhinis*, *Haemophilus parasuis*, *Streptococcus suis*, *Staphylococcus aureus*, and *Pasteurella multocida* (16,18–20,22). The association of *S. aureus* and *M. hyorhinis* with pathological lesions related to arthritis has been reported (23–25).

The objective of this study was to determine if information from abattoirs can provide an additional and reliable data source for surveillance of disease outbreaks in finisher pigs in Ontario. The validity of these data for disease surveillance was evaluated by comparing the changes in patterns in lesions among condemned carcasses with the expected changes based on the documented outbreak of PCVAD that occurred in 2005 and ended in 2007 with the introduction and widespread use of a PCV-2 vaccine. The potential changes were examined using a statistical model and a temporal cluster detection technique commonly used for disease surveillance; these 2 methods were performed to evaluate if peaks in rates observed in the scan statistics could be compared with the trends observed in negative binomial models. The temporal scan statistic is a standard method for the detection of disease clusters, and it was performed to determine its potential role in a disease surveillance system.

Materials and methods

Data source

The data, including lesion codes from January 2005 to December 2007, were obtained from the database of 1 federally inspected abattoir in Ontario. The data included total condemnations due to at least 1 of the following: abscess, arthritis, bruises, emaciation, icterus, imperfect bleeding, necrosis, nephritis, odor, over-scald, peritonitis, pleuritis, pneumonia, septicemia, toxemia, erysipelas, ascites, neoplasm, pericarditis, anemia, uremia, and enteritis. Pathological lesions that were considered to be easily identifiable, and associated with problems related to PCVAD infection (arthritis, nephritis, pneumonia, enteritis) were analyzed. Additional data recorded included: date of condemnation, total animals slaughtered, total number of condemned and total number of animals approved. The recording of lesions was performed on a daily basis by 5 to 6 veterinary meat inspectors.

Description of pathological lesions

The meat inspectors have standard criteria for classifying pathological lesions. Arthritis was associated with inflammation and presence of excessive synovial fluid in joints as well as enlarged iliac and inguinal lymph nodes. Nephritis was associated with enlargement, change in color and/or firmness of the kidneys. For arthritis and nephritis, the general appearance of the carcass was also evaluated before total condemnation. Pneumonia was associated with firm discolored lungs that could have fluid, abscesses or consolidated areas. Enteritis was associated with thickening of the intestinal walls and mega colon.

Statistical analyses

Descriptive statistics

The total number of condemned carcasses, the number condemned over the study period and by year, and rates of condemned per 10 000 animals were reported for each of the condemnation codes included in the analysis. The total and yearly condemnation rates were calculated by dividing the number of animals condemned for each specific lesion by the total number of animals entering the abattoir in specific years. This value was then multiplied by 10 000 to represent the condemnation rate per 10 000 animals.

Statistical models

Negative binomial models were used to model the effect of year and season on the rate of all condemnations consistent with PCVAD, and specific lesions associated with PCVAD. Seasons were defined as follows: winter = January, February, March; spring = April, May, June; summer = July, August, September; and fall = October, November, December. Season and year were modeled as dummy variables. Winter and 2005 were the referent variables for each model. Interactions between year and season were also examined. The model building process included the evaluation of the contribution of subsets of predictors (season, year, and interactions among these variables) in the model by using the likelihood ratio test. Nested models with and without interaction terms were compared. If the likelihood

Table 1. Numbers and rates per 10 000 hogs slaughtered over the study period and per year of total condemned and condemned due to arthritis, enteritis, nephritis, and pneumonia from a federally inspected abattoir in southern Ontario from 2005–2007

	Total	Condemned	Arthritis	Enteritis	Nephritis	Pneumonia
	Number ^a slaughtered	Number rate ^b	Number rate	Number rate	Number rate	Number rate
2005–2007	6 204 702	22 980	1911	658	1595	1255
		37	3.1	1.0	2.6	2.0
2005	1 991 726	9707	783	86	346	589
		49	3.9	0.4	1.7	2.9
2006	2 047 321	7000	610	230	516	383
		34	2.9	1.1	2.5	1.9
2007	2 165 655	6276	518	342	733	283
		29	2.4	1.6	3.4	1.3

^a Number of animals.

^b Rates per 10 000 animals slaughtered per year.

ratio test between a simpler and the full model was significant ($P < 0.05$) the interaction term was included in the model (26). In addition, models with and without the interaction terms were compared using the AIC (Akaike information criterion) and BIC (Bayesian information criterion) values. The appropriateness of using a negative binomial model, as opposed to a Poisson model, was based on the statistical significance of the over-dispersion term, alpha (26). The exposure for the negative binomial models was the total number of animals entering the abattoir each day and the outcome was the number of animals condemned for each of the condemnation codes included in this study. Anscombe and deviance residuals were calculated for each of the negative binomial models to evaluate the presence of unusual observations that would require further investigation. In addition, the normality of the Anscombe residuals was assessed to evaluate if the data fitted the model (26). In order to evaluate whether or not significant differences existed among seasons between years, contrasts were performed using the lincom command in STATA (Stata Corp., College Station, Texas, USA). This approach allows for the comparison of 2 groups based on the differences in linear combinations of the coefficients for the groups being contrasted (26). Using this approach, incidence rate ratios, standard errors, Z-scores, confidence intervals (CI), and P -values can be estimated for contrasts between 2 groups based on the model derived coefficients. The analyses of data were performed in STATA 9 (Stata Corp.). All tests performed were 2-tailed with a statistical significance level of 5%.

Temporal scan statistics

Retrospective temporal scan statistics were performed to identify the most likely temporal clusters for each condemnation code occurring from 2005 to 2007 using Bernoulli models in SaTScan Version 7.0 (27). In the temporal scan statistic, the scanning begins as a point at the smallest scale defined in the study at each point in time. Each time an additional time is reached, a likelihood ratio and the relative risk are calculated to determine if the rate of disease within each window is different from outside the window based on a Bernoulli model for these data. Monte Carlo simulations generating random replications of the dataset under the appropriate null hypothesis are used to determine the significance of the results (27).

Although we had information on the total population, the Poisson model was not used due to potential biases in estimating clusters associated with periods, such as holidays and weekends, when the abattoir was not open for slaughter, but the model would include these time periods in estimating background population levels. The Bernoulli model uses case and control data. Cases included the number of animals condemned for a specific lesion. Controls included all carcasses not condemned for the specific lesion being analyzed. All condemnation codes were analyzed separately. The maximum scanning window was restricted to 50% of the study period (2005–2007), and 9999 Monte Carlo replications were performed to estimate the statistical significance of the clusters. All scans were restricted to identify clusters with high rates of condemnations, and only the most likely significant ($P < 0.05$) cluster was reported for each condemnation code.

Validation of temporal trends

The temporal patterns observed in the negative binomial models and the significant clusters of pathological lesions found in the retrospective temporal scan statistics were qualitatively compared to PCVAD diagnostic reports from the AHL at the University of Guelph from 2005 to 2008 (Available from http://www.labservices.uoguelph.ca/labserv/units/ahl/news_notes.cfm Last accessed November 8, 2010). The AHL offers a wide range of veterinary diagnostic tests and laboratory services in various specialty areas, such as pathology, bacteriology, and virology, and it is the primary location for swine practitioners in Ontario to submit case material for diagnostic and routine monitoring services.

Results

Descriptive statistics

A total of 6 204 702 animals originating from 1132 herds were slaughtered in this abattoir from 2005–2007. A total of 22 980 pigs (37 condemnations/10 000 animals) were condemned during this period. Although the number of animals slaughtered increased from 2005 to 2007, the number and rates of total condemnations decreased. The rates of the lesions included in this study ranged between 1 to 4 condemnations/10 000 animals/y, and out of the total condemned, 5419 (23.6%)

Table 2. Negative binomial models that include the effect of season, year, and the interaction between season and year on the rate of different condemnation codes from a federally inspected abattoir in southern Ontario from 2005–2007

	Total condemned			Arthritis			Pneumonia			Enteritis			Nephritis		
	IRR ^a	95% CI ^b	P-value	IRR	95% CI	P-value	IRR	95% CI	P-value	IRR ^a	95% CI ^b	P-value	IRR	95% CI	P-value
Season ^c															
Spring	0.93	0.83–1.04	0.22	1.39	1.06–1.82	0.02	0.84	0.58–1.22	0.35	1.01	0.53–1.93	0.97	2.23	1.50–3.33	< 0.001
Summer	1.01	0.90–1.13	0.83	1.33	1.02–1.73	0.03	1.03	0.72–1.46	0.85	1.10	0.60–2.04	0.74	1.73	1.16–2.60	0.007
Fall	1.05	0.94–1.17	0.37	1.13	0.86–1.49	0.34	0.68	0.47–0.98	0.04	0.71	0.36–1.41	0.33	2.88	1.97–4.21	< 0.001
Year ^d															
2006	1.01	0.90–1.12	0.88	1.30	1.00–1.69	0.05	1.04	0.73–0.47	0.81	1.02	0.55–1.91	0.93	2.65	1.81–3.88	< 0.001
2007	0.89	0.80–0.99	0.05	1.09	0.84–1.43	0.49	0.61	0.42–0.89	0.01	7.29	4.41–12.06	< 0.001	5.48	3.82–7.84	< 0.001
Interactions															
Spring 2006	0.62	0.53–0.73	< 0.001	0.50	0.34–0.74	< 0.001	0.44	0.25–0.76	0.003	1.99	0.86–4.63	0.10	0.38	0.23–0.64	< 0.001
Spring 2007	0.86	0.73–1.01	0.06	0.70	0.48–1.01	0.06	1.27	0.75–2.16	0.36	0.51	0.25–1.07	0.07	0.45	0.29–0.72	0.001
Summer 2006	0.46	0.39–0.55	< 0.001	0.32	0.22–0.48	< 0.001	0.26	0.15–0.46	< 0.001	2.59	1.17–5.74	0.02	0.62	0.38–1.02	0.06
Summer 2007	0.39	0.39–0.46	< 0.001	0.36	0.24–0.54	< 0.001	0.29	0.16–0.53	< 0.001	0.28	0.14–0.58	0.001	0.26	0.16–0.43	< 0.001
Fall 2006	0.67	0.57–0.78	< 0.001	0.55	0.37–0.79	0.002	0.65	0.39–1.11	0.11	5.21	2.26–12.02	< 0.001	0.46	0.29–0.74	< 0.001
Fall 2007	0.46	0.39–0.53	< 0.001	0.21	0.13–0.32	< 0.001	0.29	0.15–0.55	< 0.001	0.14	0.67–0.34	< 0.001	0.13	0.87–0.20	< 0.001

^a Incidence rate ratio.^b Confidence intervals.^c Referent winter.^d Referent 2005.**Table 3.** Contrasts derived from the negative binomial models* for different condemnation codes to determine the differences among years and seasons

	Total condemned			Pneumonia			Arthritis			Nephritis			Enteritis		
	IRR ^a	95% CI ^b	P-value	IRR	95% CI	P-value	IRR	95% CI	P-value	IRR ^a	95% CI ^b	P-value	IRR	95% CI	P-value
2005 vs 2006															
Winter	0.99	0.88–1.10	0.88	0.95	0.67–1.36	0.8	0.77	0.59–0.99	0.05	0.37	0.25–0.55	< 0.001	0.97	0.52–1.81	0.94
Spring	1.59	1.41–1.79	< 0.001	2.16	1.42–3.29	< 0.001	1.51	1.15–1.99	0.003	0.97	0.70–1.33	0.84	0.48	0.27–0.86	0.01
Summer	2.12	1.89–2.37	< 0.001	3.58	2.35–5.46	< 0.001	2.35	1.75–3.15	< 0.001	0.60	0.44–0.82	0.002	0.37	0.22–0.62	< 0.001
Fall	1.47	1.32–1.64	< 0.001	1.45	0.98–2.5	0.06	1.40	1.07–1.83	0.01	0.81	0.62–1.05	0.12	0.18	0.10–0.32	< 0.001
2005 vs 2007															
Winter	1.11	1.00–1.24	0.05	1.62	1.12–2.34	0.01	0.91	0.69–1.19	0.5	0.18	0.13–0.26	< 0.001	0.14	0.08–0.22	< 0.001
Spring	1.29	1.16–1.46	< 0.001	1.27	0.87–1.86	0.21	1.30	1.00–1.69	0.04	0.39	0.30–0.52	< 0.001	0.26	0.15–0.45	< 0.001
Summer	2.81	2.49–3.16	< 0.001	5.47	3.46–8.64	< 0.001	2.50	1.86–3.33	< 0.001	0.68	0.50–0.94	0.02	0.48	0.28–0.79	0.005
Fall	2.43	2.17–2.73	< 0.001	5.50	3.28–9.2	< 0.001	4.31	3.04–6.12	< 0.001	1.41	1.06–1.89	0.02	0.93	0.47–1.84	0.84
2006 vs 2007															
Winter	1.13	1.01–1.25	0.03	1.69	1.18–2.4	0.004	1.18	0.92–1.52	0.18	0.48	0.37–0.62	< 0.001	0.14	0.08–0.20	< 0.001
Spring	0.81	0.72–0.92	< 0.001	0.58	0.39–0.89	0.01	0.86	0.65–1.13	0.28	0.41	0.31–0.54	< 0.001	0.54	0.35–0.83	0.005
Summer	1.32	1.17–1.50	< 0.001	1.52	0.90–2.57	0.11	1.06	0.76–1.47	0.73	1.13	0.84–1.51	0.39	1.27	0.84–1.93	0.25
Fall	1.65	1.47–1.85	< 0.001	3.77	2.22–6.39	< 0.001	3.08	2.15–4.41	< 0.001	1.74	1.32–2.31	< 0.001	4.99	2.95–8.45	< 0.001

* From Table 2.

^a Incidence rate ratio.^b Confidence intervals.

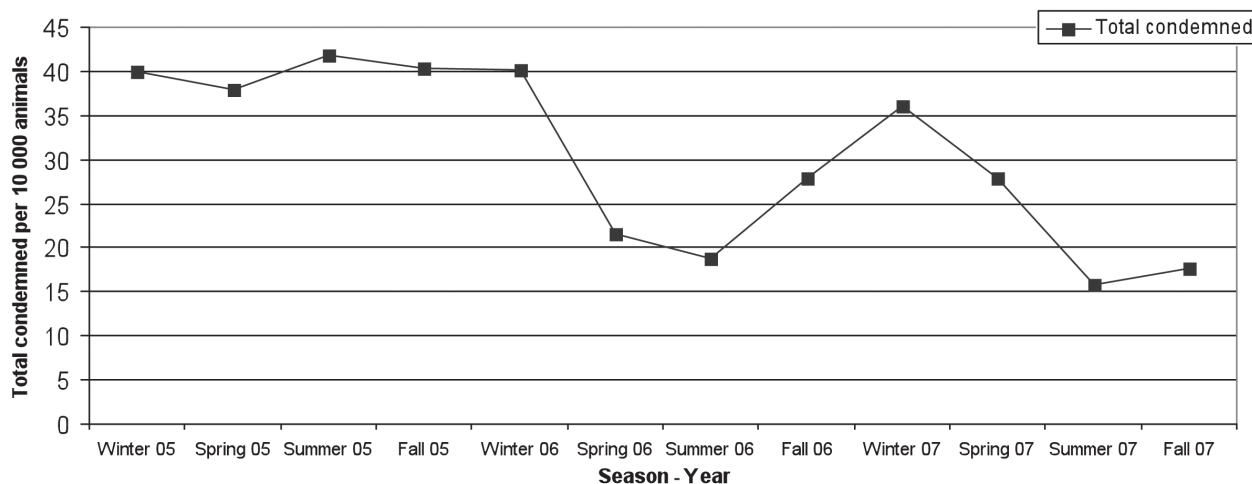


Figure 1. Total condemnation rates per 10 000 hogs slaughtered per season from a federally inspected abattoir in southern Ontario from 2005–2007.

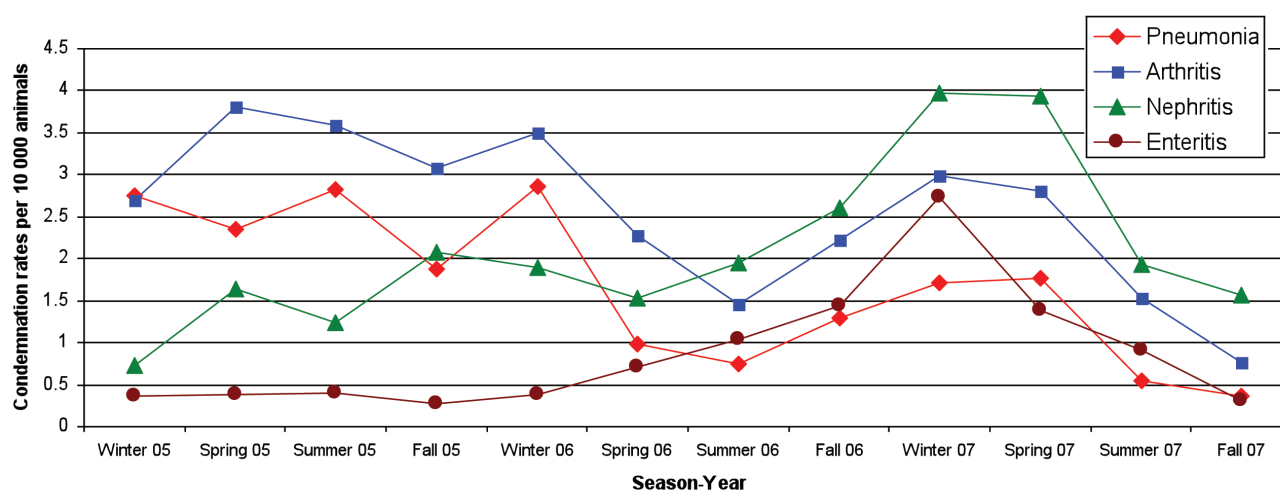


Figure 2. Condemnation rates per 10 000 hogs slaughtered per season of condemnations due to pneumonia, arthritis, nephritis, and enteritis from a federally inspected abattoir in southern Ontario from 2005–2007.

animals were condemned due to the pathological lesions commonly associated with PCVAD (Table 1). A decrease in the rates of arthritis and pneumonia was observed from 2005 to 2007. In contrast, an increase in the rates of nephritis and enteritis were observed from 2005 to 2007 (Table 1).

Statistical models

The negative binomial models showed that season, year and an interaction term for season and year were statistically significant for each model (Table 2). The likelihood ratio tests for alpha were highly significant for all the negative binomial models of each condemnation code ($P < 0.001$) indicating that overdispersion was present and negative binomial models would fit the data better than Poisson models. The Anscombe residuals showed that the data fitted the model and there were no unusual observations when plotting the deviance residuals. Contrasts were constructed from the fitted models to determine when there were significant changes in condemnation rates for the condemnation lesions included in this study (Table 3).

Total condemned

Significantly higher rates of total condemnations were observed in the winter of 2005 and 2006 compared to the winter of 2007 (Table 3; Figure 1). Significantly higher rates of total condemnations were observed in spring, summer, and fall of 2005 compared to the same seasons in 2006 and 2007 (Table 3; Figure 2).

Pneumonia

Significantly higher rates of pneumonia were observed in the spring and summer of 2005 compared to the same seasons in 2006 (Table 3; Figure 2). Significantly higher rates of pneumonia were also observed for winter, summer, and fall of 2005 compared to the same seasons in 2007 (Table 3; Figure 2). Higher rates of pneumonia were observed in 2006 compared to 2007, except for spring when a significantly lower rate of pneumonia was observed in 2006 compared to 2007, and summer when there was no significant difference between the 2 periods (Table 3; Figure 2).

Table 4. Significant most likely temporal clusters of different condemnation codes, using Bernoulli models in SaTScan from a federally inspected abattoir in Southern Ontario from 2005–2007

Most likely cluster	Total condemned	Arthritis	Pneumonia	Nephritis	Enteritis
Total population in study period	6 204 702	6 204 702	6 204 702	6 204 702	6 204 702
Total number of cases in study period	22 983	1911	1255	1595	658
Total population in the cluster	2 514 331	2 661 366	2 496 581	1 796 545	2 483 727
Time frame of cluster	2005/January/3 2006/March/29	2005/January/31 2006/June/1	2005/January/4 2006/March/28	2006/September/18 2007/July/20	2006/May/23 2007/July/31
Number of cases in the cluster	12 301	1089	777	763	505
Expected cases in the cluster	9 313.4	819.7	504.9	461.8	263.4
Observed/expected	1.32	1.32	1.54	1.65	1.92
Relative Risk	1.69	1.76	2.4	2.25	4.94
Log likelihood ratio	791.13	76.45	119.52	126.1	183.7
P-value	0.001	0.001	0.001	0.001	0.001

Arthritis

Significantly higher arthritis rates were observed in spring, summer, and fall of 2005 compared to the same seasons in 2006 and 2007 (Table 3; Figure 2). A significantly higher rate of arthritis was observed in the fall of 2006 compared to the fall of 2007 (Table 3; Figure 2).

Nephritis

Significantly lower rates of nephritis were observed in winter and summer of 2005 compared to 2006 (Table 3; Figure 2). Significantly lower rates were observed in winter, spring and summer of 2005 compared to the same seasons in 2007 (Table 3; Figure 2). However, a significantly higher rate of nephritis was observed in the fall of 2005 compared to the fall of 2007 (Table 3; Figure 2). Highly significant lower rates of nephritis were observed in the winter and spring of 2006 compared to the same seasons in 2007 (Table 3; Figure 2); however, higher rates of condemnations were observed in the fall of 2006 compared to the fall of 2007 (Table 3; Figure 2).

Enteritis

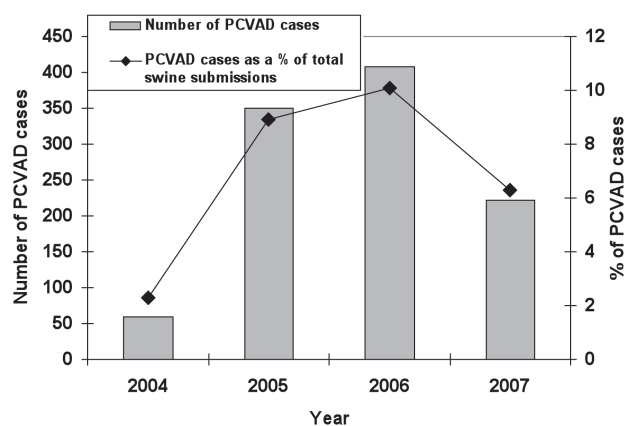
Significantly lower rates of enteritis were observed in spring, summer, and fall of 2005 compared to 2006 (Table 3; Figure 2). Significantly lower rates were observed in winter, spring, and summer of 2005 compared to the same seasons in 2007, and in the winter and spring of 2006 compared to the same seasons in 2007 (Table 3; Figure 2). However, a significantly higher rate of enteritis was observed in the fall of 2006 compared to the fall of 2007 (Table 3; Figure 2).

Temporal scan statistics

Scan statistics indicated that the most likely statistically significant temporal clusters during the entire study period for total condemned, arthritis, and pneumonia occurred at similar time frames, from the January 2005 to spring 2006 (Table 4). In contrast, the most likely statistically significant temporal clusters for the entire period for nephritis and enteritis occurred near the end of the study (Table 4).

Validation of temporal trends

The following changes in PCVAD were noted from the review of the AHL reports between 2005 and 2007:

**Figure 3.** Total number of PCVAD^a cases and total PCVAD cases as a percent of total swine submissions to the AHL^b from 2004 to 2007.

^a PCVAD – porcine circovirus-associated disease.

^b AHL – Animal Health Laboratory, University of Guelph.

- An increase in the number and proportion of PCVAD cases was observed in 2005 and 2006 compared to 2004 reflecting the outbreak of PCVAD in Ontario (Figure 3).
- A decrease in PCVAD number and proportion of cases was reported in 2007 reflecting the wide use of PCV-2 vaccine in 2007 (17) (Figure 3).
- In addition, the AHL reported a marked decline of PCV-2 diagnoses from 10.2% of the total swine submissions in the first trimester of 2006 to 8.4% by the fall of 2006 reflecting the initiation of the PCV-2 vaccine in the spring of 2006 (28).

Discussion

The higher rates of condemnation observed in 2005 in the negative binomial models and the most likely temporal clusters identified in this study for total condemned, pneumonia, and arthritis are consistent with the presence of new strains of porcine circovirus Type 2 (16,18–19). Pneumonia is one of the most frequent clinical signs observed in PCVAD (15–16). However, a wide spectrum of infectious agents such as *M. hyopneumoniae*, *M. hyorhinis*, *H. parasuis*, *S. suis*, *S. aureus*, and *P. multocida* (16,18–20,21) were found concomitant with PCV-2 on affected farms during the outbreak in Ontario. The association of these

specific pathogens such as *M. hyorhinis* and *S. aureus* with pathological lesions related to arthritis (23–25) may explain the similar patterns of condemnation rates observed for these lesions as with pneumonia. Associations between arthritis and pneumonia have been described in other studies (23); these conditions are often associated with lowered immunity which might be caused by infectious agents such as porcine circovirus, together with management and environmental stressors.

A decrease in the rate of pneumonia in the spring of 2006 and the dates when the temporal clusters identified for these condemnation codes ended coincides with a marked decline in the diagnoses of PCVAD at the AHL from 10.2% of the total swine submissions in the first months of 2006 to 8.4% by the fall of 2006 (28) likely reflecting the initiation of the use of PCV-2 vaccines in the April of 2006 in Ontario (21). In addition, a decline in the percent of positive PRRSV-PCR test results from 45% in 2005 to 33% in 2006 was also reported from the AHL that is consistent with the decline in pneumonia cases in this abattoir in 2006 (29).

The PCV-2 outbreak cases in Ontario were also associated with an increase in enteritis and nephritis (15,16). However, in contrast to what was expected, the rates of nephritis and enteritis were often higher in 2007 compared to 2005. The most common pathogens associated with a thickening of intestinal walls in slaughter pigs are *Salmonella* Typhimurium and *Lawsonia intracellularis*. These 2 pathogens are commonly found in pig populations in Ontario (30,31), but there are no records from the AHL to support an increase in the incidence of infections with these pathogens during this period. Similarly, nephritis in finisher pigs can also be caused by various conditions, and condemnations due to interstitial nephritis are still common in Canada (10,32). Recent studies report that porcine parvovirus (PPV), PCV-2, *Leptospira*, and PRRSV are common causes of nephritis in slaughter pigs (10,32–36). However, our results are inconsistent with changes in PCVAD clinical disease prevalence, and no diagnostic data exist to suggest that the other potential pathogens appeared at increasing rates during this period.

Due to the retrospective nature of our study, we were not able to support our results with serological or other sources of field data from the abattoir or the source farms, but our findings on the patterns of pneumonia and arthritis were consistent with the PCVAD outbreak that occurred in 2005 and decreased with the introduction of the PCV-2 vaccine in 2006. It also seems apparent that abattoir findings, in the case of nephritis and enteritis, may suggest areas for enhanced surveillance for unexplained changes in disease patterns that were not identified through traditional passive surveillance involving diagnostic laboratories. Identifying the causal agents behind these changing patterns could have considerable economic, public health, and animal health benefits. Systems such as the Pig Health Monitoring Scheme in Australia, the Health Production Surveillance System (HEPS), and PigMON in the USA were developed to inform producers and veterinarians of the presence and impact of disease by using slaughter inspection findings (1,9,37). The feasibility of using feedback from abattoirs has also been investigated in Canada (38) and various projects have attempted to implement this type of program so that slaughterhouse lesions might be

used in herd health programs (39). However, these programs have not been well-utilized by producers. This information would be of much more value if slaughter data were incorporated into a surveillance program and validated with additional health data sources to monitor the whole industry rather than relying on individual producers utilizing the information for their own herd health programs.

We were only able to receive data from 1 of 5 federally inspected plants in Ontario, and this plant also received a small percentage of animals (approximately 1%) from Quebec. In addition to the 5 federally inspected plants there are more than 200 provincially licensed plants that slaughter hogs in Ontario. Although both federal and provincial plants have strict regulated inspection standards, the federally inspected abattoirs are the only ones that can sell outside the province and the country, and their inspection practices have to meet trade requirements (40). Consequently, federal abattoirs may tend to handle more uniform animals with higher health status, compared to provincial plants which are restricted to more local markets. Although federal data appeared to show important disease trends, we think comparisons with provincial data may also be useful because in the case of a disease outbreak, pathological problems may be reflected more markedly in provincially inspected abattoirs. Based on Food Safety Decision Support System data the condemnation rates in provincial abattoirs can be higher (41). It would be important for future studies to compare the relative utility of provincial versus federal abattoir data for disease surveillance based on the quality of the data, its consistency with known disease patterns, and the ability to make inferences to specific spatial locations. The availability of spatial information that reflected the general location of affected farms would enhance the use of these data for surveillance. However, the impact of variation in inspection quality among abattoirs and stable spatial variation in condemnation rates due to environmental and management conditions should be considered when integrating data from several abattoirs (24,42).

Acknowledgments

We thank the abattoir plant for providing the data used in this study. The authors also thank the Ontario Ministry of Agriculture, Food, and Rural Affairs (OMAFRA); the Agriculture and Agri-Food Canada, Food Safety Initiative (FSI); and the Animal Health Strategic Investment (AHSI) for funding. CVJ

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